RPractical_Divinagracia

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PART A: Basic Operations A1

```
v<-seq(5,50, by = 5)

print(v)
```

```
##  [1]  5 10 15 20 25 30 35 40 45 50
```

```
print(length(v))
```

```
## [1] 10
```

```
print(sum(v))
```

```
## [1] 275
```

```
print(mean(v))
```

```
## [1] 27.5
```

A2

```
v <-rep(c(1,2,3), times = 4)
z <-rep(c(1,2,3), times = 4)
print(v)
```

```
##  [1] 1 2 3 1 2 3 1 2 3 1 2 3
```

```
print(z)
```

```
##  [1] 1 2 3 1 2 3 1 2 3 1 2 3
```

A3

```
print(17 %%5)
```

```
## [1] 2
```

```
print(17 %/%5)
```

```
## [1] 3
```

```
print(2*10)
```

```
## [1] 20
```

```
print(sqrt(144))
```

```
## [1] 12
```

```r
print(round(pi,3))
```

```
## [1] 3.142
```

A4

```r
a<-c(2,4,6,8)
b<-c(1,3,5,7)
print(a + b)
```

```
## [1]  3  7 11 15
```

```r
print(a * b)
```

```
## [1]  2 12 30 56
```

```r
print(a > b)
```

```
## [1] TRUE TRUE TRUE TRUE
```

```r
print(sum(a * b))
```

```
## [1] 100
```

A5

```r
students <-paste("Student-", 1:3,sep="")
print(students)
```

```
## [1] "Student-1" "Student-2" "Student-3"
```

```r
print(typeof(42))
```

```
## [1] "double"
```

```r
print(typeof(42L))
```

```
## [1] "integer"
```

```r
print(typeof("42"))
```

```
## [1] "character"
```

```r
print(typeof(4i))
```

```
## [1] "complex"
```

```r
print(typeof(TRUE))
```

```
## [1] "logical"
```

PART B: Data Frames and Conditional Statements

B1

```r
students <- data.frame(id = 1:8,
name= c("Alice","Bruno","Chen","Dana","Eli","Farid","Grae","Hiro"),
sex = c("F","M","M","F","F","M","F","M"),
section = c("A","B","A","C","B","C","A" ,"B"),
midterm = c(78,85,92,67,74,88,95,59),
final= c(82,79,96,71,80,91,93,65),
attendance =c(95,88,100,72,90,85,98,60),
```

```
stringsAsFactors = FALSE)
print(students)
```

```
##   id  name sex section midterm final attendance
## 1  1 Alice   F       A      78    82         95
## 2  2 Bruno   M       B      85    79         88
## 3  3  Chen   M       A      92    96        100
## 4  4  Dana   F       C      67    71         72
## 5  5   Eli   F       B      74    80         90
## 6  6 Farid   M       C      88    91         85
## 7  7  Grae   F       A      95    93         98
## 8  8  Hiro   M       B      59    65         60
```

B2

```
str(students)
```

```
## 'data.frame':    8 obs. of  7 variables:
##  $ id        : int  1 2 3 4 5 6 7 8
##  $ name      : chr  "Alice" "Bruno" "Chen" "Dana" ...
##  $ sex       : chr  "F" "M" "M" "F" ...
##  $ section   : chr  "A" "B" "A" "C" ...
##  $ midterm   : num  78 85 92 67 74 88 95 59
##  $ final     : num  82 79 96 71 80 91 93 65
##  $ attendance: num  95 88 100 72 90 85 98 60
```

```
dim(students)
```

```
## [1] 8 7
```

```
names(students)
```

```
## [1] "id"         "name"       "sex"        "section"    "midterm"
## [6] "final"      "attendance"
```

```
summary(students)
```

```
##        id              name               sex             section     
##  Min.   :1.00   Length   :8   Length   :8   Length   :8   Min.   :59.00
##  1st Qu.:2.75   N.unique :8   N.unique :2   N.unique :3   1st Qu.:72.25
##  Median :4.50   N.blank  :0   N.blank  :0   N.blank  :0   Median :81.50
##  Mean   :4.50   Min.nchar:3   Min.nchar:1   Min.nchar:1   Mean   :79.75
##  3rd Qu.:6.25   Max.nchar:5   Max.nchar:1   Max.nchar:1   3rd Qu.:89.00
##  Max.   :8.00                                            Max.   :95.00
##      final         attendance
##  Min.   :65.00   Min.   : 60.00
##  1st Qu.:77.00   1st Qu.: 81.75
##  Median :81.00   Median : 89.00
##  Mean   :82.12   Mean   : 86.00
##  3rd Qu.:91.50   3rd Qu.: 95.75
##  Max.   :96.00   Max.   :100.00
```

B3

```
students$average <- round(rowMeans(students[, c("midterm", "final")]),1)
print(students)
```

```
##   id  name sex section midterm final attendance average
## 1  1 Alice   F       A      78    82         95    80.0
```

```
## 2  2 Bruno   M       B      85    79         88    82.0
## 3  3  Chen   M       A      92    96        100    94.0
## 4  4  Dana   F       C      67    71         72    69.0
## 5  5   Eli   F       B      74    80         90    77.0
## 6  6 Farid   M       C      88    91         85    89.5
## 7  7  Grae   F       A      95    93         98    94.0
## 8  8  Hiro   M       B      59    65         60    62.0
```

B4

```
students$passed <-
  students$average >= 70 & students$attendance >= 75
print(students)
```

```
##   id  name sex section midterm final attendance average passed
## 1  1 Alice   F       A      78    82         95    80.0   TRUE
## 2  2 Bruno   M       B      85    79         88    82.0   TRUE
## 3  3  Chen   M       A      92    96        100    94.0   TRUE
## 4  4  Dana   F       C      67    71         72    69.0  FALSE
## 5  5   Eli   F       B      74    80         90    77.0   TRUE
## 6  6 Farid   M       C      88    91         85    89.5   TRUE
## 7  7  Grae   F       A      95    93         98    94.0   TRUE
## 8  8  Hiro   M       B      59    65         60    62.0  FALSE
```

B5

```
students$standing <-
ifelse(students$average>=90,"Honors",
ifelse(students$average>= 75, "Good", "Needs Support"))

print(students)
```

```
##   id  name sex section midterm final attendance average passed      standing
## 1  1 Alice   F       A      78    82         95    80.0   TRUE          Good
## 2  2 Bruno   M       B      85    79         88    82.0   TRUE          Good
## 3  3  Chen   M       A      92    96        100    94.0   TRUE        Honors
## 4  4  Dana   F       C      67    71         72    69.0  FALSE Needs Support
## 5  5   Eli   F       B      74    80         90    77.0   TRUE          Good
## 6  6 Farid   M       C      88    91         85    89.5   TRUE          Good
## 7  7  Grae   F       A      95    93         98    94.0   TRUE        Honors
## 8  8  Hiro   M       B      59    65         60    62.0  FALSE Needs Support
```

B6

```
classify <- function(score){
  if (score >= 85){
    return("Distinction")
  }else if (score>= 70){
    return("Merit")
  }else if (score>=50){
    return("Pass")
  }else{
    return("Fail")

  }
}
result<-sapply(c(90,73,55,40),classify)
```

```r
print(result)
```

```
## [1] "Distinction" "Merit"       "Pass"        "Fail"
```

Part C: rbind() and cbind()

C1

```r
new_students <- data.frame(
  id = c(9, 10),
  name = c("Ines", "Jamal"),
  sex = c("F", "M"),
  section = c("C", "A"),
  midterm = c(81, 70),
  final = c(84, 68),
  attendance = c(92, 80),
  average = c(87, 69),
  passed = c(TRUE, FALSE),
  standing = c("Good", "Needs Support")
)
names(students)
```

```
##  [1] "id"         "name"       "sex"        "section"    "midterm"
##  [6] "final"      "attendance" "average"    "passed"     "standing"
```

```r
str(students)
```

```
## 'data.frame':    8 obs. of  10 variables:
##  $ id        : int  1 2 3 4 5 6 7 8
##  $ name      : chr  "Alice" "Bruno" "Chen" "Dana" ...
##  $ sex       : chr  "F" "M" "M" "F" ...
##  $ section   : chr  "A" "B" "A" "C" ...
##  $ midterm   : num  78 85 92 67 74 88 95 59
##  $ final     : num  82 79 96 71 80 91 93 65
##  $ attendance: num  95 88 100 72 90 85 98 60
##  $ average   : num  80 82 94 69 77 89.5 94 62
##  $ passed    : logi  TRUE TRUE TRUE FALSE TRUE TRUE ...
##  $ standing  : chr  "Good" "Good" "Honors" "Needs Support" ...
```

```r
all_students <- rbind(students, new_students)

print(all_students)
```

```
##    id  name sex section midterm final attendance average passed      standing
## 1   1 Alice   F       A      78    82         95    80.0   TRUE          Good
## 2   2 Bruno   M       B      85    79         88    82.0   TRUE          Good
## 3   3  Chen   M       A      92    96        100    94.0   TRUE        Honors
## 4   4  Dana   F       C      67    71         72    69.0  FALSE Needs Support
## 5   5   Eli   F       B      74    80         90    77.0   TRUE          Good
## 6   6 Farid   M       C      88    91         85    89.5   TRUE          Good
## 7   7  Grae   F       A      95    93         98    94.0   TRUE        Honors
## 8   8  Hiro   M       B      59    65         60    62.0  FALSE Needs Support
## 9   9  Ines   F       C      81    84         92    87.0   TRUE          Good
## 10 10 Jamal   M       A      70    68         80    69.0  FALSE Needs Support
```

C2

```r
quiz <- c(8, 7, 10, 6, 7, 9, 10, 5, 8, 6)

all_students <- cbind(all_students, quiz = quiz)

print(class(all_students))
```

```
## [1] "data.frame"
```

```r
print(all_students)
```

```
##    id  name sex section midterm final attendance average passed      standing
## 1   1 Alice   F       A      78    82         95    80.0   TRUE          Good
## 2   2 Bruno   M       B      85    79         88    82.0   TRUE          Good
## 3   3  Chen   M       A      92    96        100    94.0   TRUE        Honors
## 4   4  Dana   F       C      67    71         72    69.0  FALSE Needs Support
## 5   5   Eli   F       B      74    80         90    77.0   TRUE          Good
## 6   6 Farid   M       C      88    91         85    89.5   TRUE          Good
## 7   7  Grae   F       A      95    93         98    94.0   TRUE        Honors
## 8   8  Hiro   M       B      59    65         60    62.0  FALSE Needs Support
## 9   9  Ines   F       C      81    84         92    87.0   TRUE          Good
## 10 10 Jamal   M       A      70    68         80    69.0  FALSE Needs Support
##    quiz
## 1     8
## 2     7
## 3    10
## 4     6
## 5     7
## 6     9
## 7    10
## 8     5
## 9     8
## 10    6
```

C3

```r
m <- cbind(
  midterm = all_students$midterm,
  final = all_students$final
)

mean_row <- colMeans(m)

m <- rbind(m, mean = mean_row)

print(m)
```

```
##      midterm final
##         78.0  82.0
##         85.0  79.0
##         92.0  96.0
##         67.0  71.0
##         74.0  80.0
##         88.0  91.0
##         95.0  93.0
##         59.0  65.0
##         81.0  84.0
```

```
##         70.0  68.0
## mean    78.9  80.9
```

C4

```
bad <- data.frame(wrong_column = 1)

tryCatch(
  rbind(students, bad),
  error = function(e) {
    message("rbind failed: ", conditionMessage(e))
    NULL
  }
)
```

```
## rbind failed: numbers of columns of arguments do not match
```

```
## NULL
```

```
# rbind() may produce an error when the data frames have
# different column names or incompatible structures.
# tryCatch() safely catches the error and displays an error
# message instead of stopping the entire R program.
```

Part D:Subsetting D1

```
print(all_students$name)
```

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Eli"   "Farid" "Grae"  "Hiro"  "Ines"
## [10] "Jamal"
```

```
print(all_students[["name"]])
```

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Eli"   "Farid" "Grae"  "Hiro"  "Ines"
## [10] "Jamal"
```

```
print(all_students[, "name"])
```

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Eli"   "Farid" "Grae"  "Hiro"  "Ines"
## [10] "Jamal"
```

D2

```
print(all_students[2:5, c("name", "midterm", "final")])
```

```
##    name midterm final
## 2 Bruno      85    79
## 3  Chen      92    96
## 4  Dana      67    71
## 5   Eli      74    80
```

D3

```
print(all_students[all_students$attendance > 90, ])
```

```
##   id  name sex section midterm final attendance average passed standing quiz
## 1  1 Alice   F       A      78    82         95      80   TRUE     Good    8
## 3  3  Chen   M       A      92    96        100      94   TRUE   Honors   10
## 7  7  Grae   F       A      95    93         98      94   TRUE   Honors   10
## 9  9  Ines   F       C      81    84         92      87   TRUE     Good    8
```

D4

```
print(subset(
  all_students,
  sex == "F" &
    section %in% c("A", "C") &
    final >= 75,
  select = c(name, section, final)
))
```

```
##    name section final
## 1 Alice       A    82
## 7  Grae       A    93
## 9  Ines       C    84
```

D5

```
print(all_students$name[which.max(all_students$average)])
```

```
## [1] "Chen"
```

```
print(all_students$name[which.min(all_students$average)])
```

```
## [1] "Hiro"
```

D6

```
sorted_students <- all_students[
  order(all_students$average, decreasing = TRUE),
]

print(head(sorted_students, 3))
```

```
##   id  name sex section midterm final attendance average passed standing quiz
## 3  3  Chen   M       A      92    96        100    94.0   TRUE   Honors   10
## 7  7  Grae   F       A      95    93         98    94.0   TRUE   Honors   10
## 6  6 Farid   M       C      88    91         85    89.5   TRUE     Good    9
```

Part E: Basic factor E1

```
all_students$section <- factor(all_students$section)
all_students$sex <- factor(all_students$sex)

print(levels(all_students$section))
```

```
## [1] "A" "B" "C"
```

```
print(levels(all_students$sex))
```

```
## [1] "F" "M"
```

E2

```
print(table(all_students$sex))
```

```
##
## F M
## 5 5
```

```
print(table(all_students$section))
```

```
##
## A B C
```

```
## 4 3 3
```

```
print(table(all_students$sex, all_students$section))
```

```
##
##     A B C
##   F 2 1 2
##   M 2 2 1
```

E3

```
print(as.integer(all_students$sex))
```

```
##  [1] 1 2 2 1 1 2 1 2 1 2
```

```
# The integers represent the internal numeric codes
# assigned to each factor level.
# They do not represent the actual sex values.
```

E4

```
all_students$sex_label <- factor(
  all_students$sex,
  levels = c("F", "M"),
  labels = c("Female", "Male")
)

print(levels(all_students$sex_label))
```

```
## [1] "Female" "Male"
```

E5

```
print(summary(all_students$sex_label))
```

```
## Female   Male
##      5      5
```

```
print(tapply(
  all_students$average,
  all_students$sex_label,
  mean
))
```

```
## Female   Male
##   81.4   79.3
```

R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see http://rmarkdown.rstudio.com.

When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

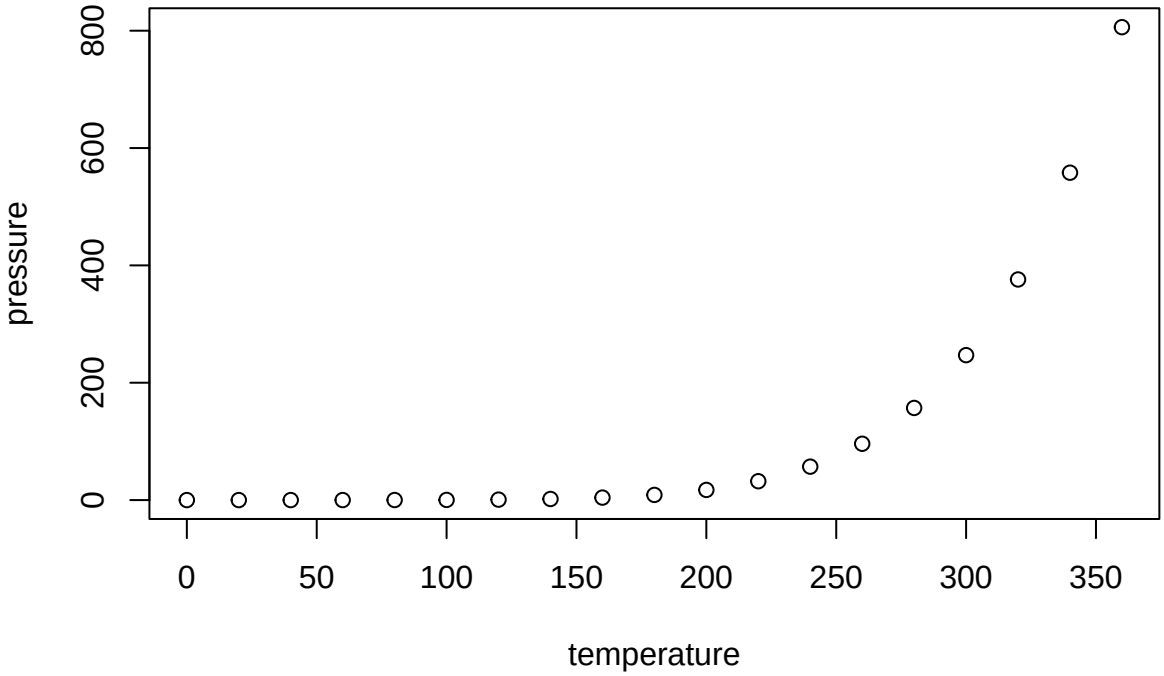
```
summary(cars)
```

```
##      speed           dist
##  Min.   : 4.0   Min.   :  2.00
##  1st Qu.:12.0   1st Qu.: 26.00
##  Median :15.0   Median : 36.00
```

```
##  Mean   :15.4   Mean   : 42.98
##  3rd Qu.:19.0   3rd Qu.: 56.00
##  Max.   :25.0   Max.   :120.00
```

Including Plots

You can also embed plots, for example:



Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.